1. Run report: EXP-ECDLP-502ae2 (RUN-ECDLP-502ae2-01)

Observations only. No verdict or support/weaken language; the evidence review is in `reports/review-*/`.

- **Experiment** EXP-ECDLP-502ae2 (instrument replication of EXP-ECDLP-f49f75 at m = 3) · **hypothesis** H-ECDLP-60de36 · **question** RQ-ECDLP-001 · **goal** GOAL-ECDLP-001
- **Run** RUN-ECDLP-502ae2-01: started 2026-10-15 16:34 UTC, finished 18:26 UTC, exit 0, 30 instances; launcher commit 43f93787df (clean tree at launch).
- **Command** `python3 experiments/EXP-ECDLP-502ae2/msolve_grid.py --plan experiments/EXP-ECDLP-502ae2/trial-plan.json --out …/RUN-ECDLP-502ae2-01/raw-result.jsonl --workers 4`; seed string `EXP-ECDLP-f49f75/v1` (instances identical to EXP-ECDLP-f49f75 RUN-02 for L = 3..5).
- **Host** 4 vCPU Intel Xeon 2.10 GHz, 15 GB; msolve 0.6.5-1build2 (1 thread per process); Singular 4.3.2 used only to expand S_4 (untimed).

1.1. Instrument and arms

Per Gröbner arm and instance: `msolve -g 2 -v 2` (reduced DRL basis; wall time; max F4 degree from the verbose log; basis used for the planted check), then, if it finished, `msolve -P 1` (full solve: F4 + FGLM parametrization; wall time). An instance counts as completed only when both finish within 600 s each. Arms: A chain encoding ($3L$ variables), B same set dense (3 variables), C random set dense, E meet-in-the-middle enumeration. Stopping rule: an arm with every instance timed out in a cell is not run at larger L.

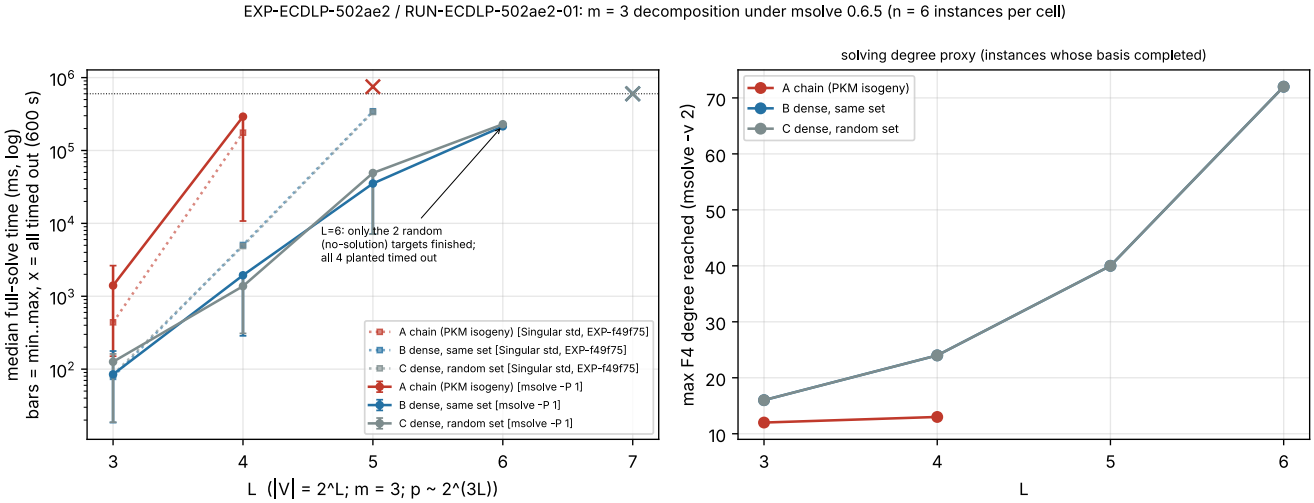


Figure 1: Left: median full-solve time (msolve `-P 1`, measured, ms) per arm vs L, bars min–max over completed instances; dotted: Singular std+fglm medians on the same instances (EXP-ECDLP-f49f75 RUN-02); × = every instance timed out. At $L = 6$ only the two random targets (no decomposition, quotient dimension 0) completed for B and C. Right: maximum F4 degree reached in the `-g 2` run, over instances whose basis completed. n = 6 per cell.

1.2. Per-cell measurements

Columns: completed / timed out / crashed / degenerate target; median full-solve ms [min–max]; median basis-only (`-g 2`) ms; max F4 degree (basis-completed instances); instances whose basis completed; planted check passed/checked.

L	p bits	arm	ok/to/crash/deg	solve ms [min–max]	gb ms	max F4 deg	gb done	planted
3	10	A	6/0/0/0	1,403 [150–2,632]	205	12	6	4/4
3	10	B	6/0/0/0	85 [18–177]	26	16	6	4/4
3	10	C	6/0/0/0	126 [19–162]	24	16	6	4/4
3	10	E	6/0/0/0	54 group ops (FB 4)	—	—	—	—
4	13	A	6/0/0/0	291,249 [10,769–295,409]	10,488	13	6	4/4
4	13	B	6/0/0/0	1,941 [286–2,049]	343	24	6	4/4
4	13	C	6/0/0/0	1,375 [309–1,960]	322	24	6	4/4
4	13	E	6/0/0/0	152 group ops (FB 8)	—	—	—	—
5	16	A	0/6/0/0	—	—	—	—	—
5	16	B	6/0/0/0	35,222 [7,126–49,713]	7,588	40	6	4/4
5	16	C	6/0/0/0	49,120 [7,106–50,496]	7,736	40	6	4/4
5	16	E	6/0/0/0	629 group ops (FB 17)	—	—	—	—

6	19	B	2/4/0/0	214,509 [213,622–215,396]	223,386	72	6	0/0
6	19	C	2/4/0/0	229,398 [227,944–230,853]	218,344	72	6	0/0
6	19	E	6/0/0/0	2,146 group ops (FB 32)	—	—	—	—
7	22	B	0/6/0/0	—	—	—	—	—
7	22	C	0/6/0/0	—	—	—	—	—
7	22	E	6/0/0/0	8,000 group ops (FB 62)	—	—	—	—

1.3. Operational notes (facts)

- Stopping rule retired arm A at $L = 5$ (all 6 instances timed out in the basis step) and arms B, C at $L = 7$.
- At $L = 6$, all 6 B and C bases completed (max F4 degree 72); the full solve timed out on the 4 planted instances and finished on the 2 random targets.
- Arm A at $L = 4$: basis step about 10 s on every instance; full solve 10.8–295 s.
- msolve 0.6.5 segfaults on the dense $m = 2$, $L = 8$ system (degree-256 univariates), so $m = 2$ was out of scope. Its input parser keeps only the last of several constant terms; the launcher always emits one merged constant.